

Multiple Choice Questions

Here are 25 basic multiple choice questions from the biology lab manual, with the correct option in bold:

1. What is the concentration term used when the solute quantity is expressed in moles per litre of solution?

- A) Molality
- B) Normality
- C) Molarity
- D) Mass percentage
- **C) Molarity****

2. Which of the following is NOT a type of solution based on solvent saturation?

- A) Saturated
- B) Supersaturated
- C) Dilute
- **D) Unsaturated****

3. Iodine test can detect the presence of:

- A) Monosaccharides
- B) Disaccharides
- **C) Polysaccharides****
- D) Lipids

4. Which reagents are required for performing Benedict's test?

- A) Iodine and sulfuric acid
- B) Ninhydrin and acetone
- C) Sodium citrate and copper sulfate
- **D) Molisch's reagent and acetic acid****

5. The Biuret test gives a color change to _____ in the presence of proteins.

- A) Yellow
- B) Blue black
- C) Brown red
- **D) Violet****

6. In paper chromatography, the R_f value indicates _____.

- A) Distance travelled by the solvent
- B) Size of the separated zones
- C) Time taken for separation
- **D) Movement of substance relative to the solvent front****

7. What are the products formed when triglycerides undergo alkaline hydrolysis?

- **A) Glycerol and soap****
- B) Fatty acids and esters

C) Peptides and glycerol

D) Glucose and fructose

8. Which enzyme present in saliva breakdown starch into maltose?

A) Protease

****B) Amylase****

C) Lipase

D) Nuclease

9. What causes a permanent loss of enzyme structure and function?

A) Inhibition

B) Denaturation

C) Competitive binding

****D) Change in pH****

10. Saponification value is defined as _____.

A) Volume of KOH required per unit mass of fat

B) Moles of glycerol released per gram of fat

C) Moles of soap formed per litre

****D) Milligrams of KOH required to neutralize fatty acids released from 1 gram of fat****

11. Ninhydrin test gives coloration for amino acids due to reaction with _____.

****A) Amino group****

B) Carboxyl group

C) Hydrogen atoms

D) R groups

12. What is the primary structure of proteins related to?

****A) Amino acid sequence****

B) Pleated sheets

C) 3D shape

D) Binding sites

13. Molisch's test confirms presence of _____ in a sample.

A) Proteins

B) Lipids

****C) Carbohydrates****

D) Vitamins

14. Iodine solution is added in Benedict's test to _____.

A) Reduce copper ions

B) Oxidize reducing sugars

****C) Detect presence of starch****

D) Make alkaline conditions

15. During paper chromatography, the solvent _____.

- A) Hydrolyzes amino acids
- B) Produces ninhydrin color
- **C) Moves biomolecules up the paper****
- D) Denatures proteins

16. Lipids produce a translucent spot in _____ test.

- **A) Spot****
- B) Solubility
- C) Iodine
- D) Molisch's

17. The Biuret reagent contains _____.

- A) Ninhydrin
- **B) Copper sulfate****
- C) Benedict's mixture
- D) Sodium carbonate

18. Xanthoproteic test detects which amino acids?

- A) Alanine
- B) Glycine
- **C) Tyrosine and tryptophan****
- D) Asparagine

19. Glycogen gives _____ color with iodine.

A) Violet

****B) Red****

C) Blue

D) Yellow

20. In chromatography, the retardation factor R_f is calculated as:

A) Distance travelled by biomolecule / Total distance

B) Total distance / Distance travelled by biomolecule

C) Distance travelled by biomolecule x Distance travelled by solvent

****D) Distance travelled by biomolecule / Distance travelled by solvent****

21. Molisch's test results in the formation of _____ complex.

****A) Furfural****

B) Copper sulfate

C) Ninhydrin

D) Biuret

22. Among lipids, steroids have _____.

****A) Low saponification values****

B) High iodine numbers

C) Low boiling points

D) High Rf values

23. Optimum pH indicates _____.

A) Highest enzyme concentration

****B) Maximum enzyme activity****

C) Minimum inhibition

D) Maximum substrate binding

24. In paper chromatography, the stationary phase is _____.

****A) Filter paper****

B) Butan-1-ol

C) Solvent mixture

D) Beaker

25. Burette in saponification value estimation is used for _____.

****A) KOH standardization****

B) Titration with HCl

C) Making standard solution

D) Dilution of fatty acids

Here are 25 conceptual multiple choice questions from the biology lab manual, with the correct option in bold:

1. During paper chromatography, separation occurs due to difference in _____ of amino acids.

A) Size

B) Charge

****C) Side chains****

D) Solubility

2. Which condition can cause protein denaturation?

A) High substrate concentration

B) Competitive inhibition

C) Optimum pH

****D) Extreme temperatures****

3. Among various solution types, supersaturated solutions are _____.

A) Very dilute

B) Saturated with solute

C) Containing maximum dissolved solute

****D) Unstable****

4. The saponification reaction leads to formation of _____.

****A) Salt and ester****

B) Soap and water

C) Fatty acid and alcohol

D) Glycerol and soap

5. Biuret test can detect the presence of _____.

A) Monosaccharides

****B) Peptides****

C) Aromatic amino acids

D) Saturated fats

6. During Benedict's test for reducing sugars, color change occurs due to _____.

A) Oxidation of aldehyde groups

B) Reduction of copper ions

****C) Formation of halide ions****

D) Reaction with molisch reagent

7. Iodine solution is used in testing for _____.

A) Proteins

B) Reducing sugars

****C) Polysaccharides****

D) Amino acids

8. Molality expresses concentration in units of _____.

A) Moles per cubic decimeter

B) Grams per cubic meter

****C) Moles per kilogram****

D) grams per liter

9. The Biuret reagent contains _____ which forms a colored complex with peptides.

****A) Copper ions****

B) Silver nitrate

C) Iodine

D) Benedict's mixture

10. Chromatography depends on interactions with _____ for separation.

A) indicator dyes

****B) stationary phase****

C) reagents

D) spectrophotometer

11. During saponification reaction, triglycerides are broken down in the presence of _____.

****A) Base****

B) Acid

C) Oxidase

D) Ligase

12. Xanthoproteic test involves adding _____ to detect proteins.

A) Biuret reagent

B) Copper sulfate

****C) Nitric acid****

D) Benedict's solution

13. The optimum pH indicates _____.

A) Highest substrate binding

- B) Minimum enzyme inhibition
- **C) Maximum enzyme activity****
- D) Complete saturation by substrate

14. The mobile phase in chromatography refers to _____.

- **A) Solvent****
- B) Filter paper
- C) Ninhydrin
- D) Beaker

15. Amino acids show differential migration during chromatography due to _____.

- A) Size differences
- **B) Varying side chains****
- C) Hydrogen bonding
- D) pH changes

16. An unsaturated solution is one which can _____.

- **A) Dissolve more solute****
- B) Precipitate solute

C) Resist temperature changes

D) Conduct electricity

17. The Biuret reaction demonstrates that _____.

A) Lipids form soaps

****B) Peptides form colored complexes****

C) Sugars are reducing agents

D) Vitamins reduce copper

18. Iodine solution causes color change with _____.

A) Amino acids

****B) Starch****

C) Cellulose

D) Glycerol

19. Saponification value helps determine _____.

A) Purity of oil

****B) Average chain length****

C) Enzyme activity

D) Protein content

20. Molisch's test confirms presence of _____.

- A) Unsaturation
- B) Amino groups
- **C) Carbohydrates****
- D) Peptide bonds

21. Ninhydrin spray is required in chromatography for _____.

- A) Denaturing proteins
- B) Visualizing separation
- **C) Detecting amino acids****
- D) Drying the paper

22. During paper chromatography, the solvent _____.

- A) Hydrolyzes macromolecules
- B) Converts amino acids into sugars
- C) Cleaves glycosidic bonds
- **D) Moves up the paper****

23. Benedict's test can detect _____.

****A) Reducing sugars****

B) Saturated fats

C) Ketones

D) Aromatic amines

24. The retardation factor R_f depends on _____.

****A) Nature of amino acid side chain****

B) Number of peptide bonds

C) Speed of solvent movement

D) Amount of protein spotted

25. Lipase enzyme acts on _____.

A) Starch

B) Glycogen

****C) Triglycerides****

D) Cellulose

Here are 25 very tough multiple choice questions from the biology lab manual, covering additional concepts. The correct option is in bold font:

1. For an enzyme exhibiting a K_m of 0.5 mM and a V_{max} of 25 $\mu\text{mol/min}$, 10 mM of substrate would yield a reaction velocity of:

A) 10 $\mu\text{mol/min}$

B) 25 $\mu\text{mol/min}$

****C) 20 $\mu\text{mol/min}$ ****

D) 15 $\mu\text{mol/min}$

2. If 10 mL of 0.02 N NaOH is required for saponification of 2 g oil sample while 9 mL is needed for blank, saponification value is:

A) 310 mg/g

B) 56 mg/g

C) 280 mg/g

****D) 250 mg/g****

3. During protein purification by ion-exchange chromatography, elution is done by changing:

****A) pH****

B) Substrate

C) Solvent polarity

D) Flow rate

4. The isoelectric point (pI) corresponds to:

- A) Maximal protein charge
- **B) No net charge on protein****
- C) Minimal buffering capacity
- D) Maximum protein stability

5. Which of the following statements about Michaelis-Menten kinetics is INCORRECT?

- **A) It applies to zero order reactions****
- B) Gives hyperbolic curve for velocity versus substrate
- C) Allows estimation of K_m and V_{max}
- D) Based on enzyme-substrate binding

6. Tyndall effect observed by passage of light beam through colloidal solution is due to:

- A) Absorbance by particles
- B) Color development
- **C) Scattering effect of suspended particles****
- D) Fluorescence

7. Molarity of HCl required to prepare 100 mL of buffer at pH 5 from 0.2 M sodium acetate and 0.2 M acetic acid is:

****A) 0.004 M****

B) 0.05 M

C) 0.3 M

D) 0.8 M

8. Which statement regarding membrane transport is CORRECT?

A) Simple diffusion needs energy

B) Osmosis depends on solute concentration gradient

****C) Electrogenic pumping generates voltage****

D) symport leads to leakage of solutes

9. Identify the effect that results from buffering capacity?

A) Solubility changes

B) Molality changes

****C) Resistance to pH change****

D) Osmolarity changes

10. Which amino acid is NOT correctly matched to its property?

- A) Tryptophan - Large hydrophobic R group
- B) Histidine - Basic polar R group capable of protonation
- C) Methionine - Nucleophilic sulfur R group
- **D) Aspartate - Aromatic polar R group****

11. Constant interaction of enzyme with inhibitors results in decrease of apparent:

- **A) V_{max} ****
- B) K_{cat}
- C) K_m
- D) Denaturation temperature

12. During lipid extraction from tissue by the Folch's method, washing with saline is crucial to remove:

- A) Phospholipids
- B) Micelles
- **C) Water soluble contaminants****
- D) Non-polar lipids

13. Hill coefficient provides information about:

- A) Degree of inhibition
- B) Temperature stability
- **C) Cooperativity during ligand binding****
- D) Effect of pH

14. Identify the property NOT shown by B-form of DNA:

- A) Right handed helix
- B) Glycosidic bonds in anti conformation
- C) Double helical structure
- **D) Parallel strands****

15. In gel filtration chromatography, separation depends MOSTLY on:

- **A) Molecular size****
- B) Charge interaction with resin
- C) Differential solvent flow
- D) Attraction to resin ligands

16. Rate of diffusion across artificial lipid membrane depends on:

- A) Temperature gradient
- **B) Lipid fluidity****

- C) Osmotic pressure
- D) Membrane asymmetry

17. Which statement about myoglobin is INCORRECT?

- A) Oxygen binding protein
- B) Found in muscle cells
- C) Contains iron as prosthetic group
- **D) Kinetics follow Henri-Michaelis model****

18. In Born-Haber cycle, lattice energy is calculated from:

- **A) Enthalpies of other reactions****
- B) Solvation energy
- C) Measurement of bond angles
- D) X-ray crystallography

19. Identify effect of competitive inhibition on Lineweaver-Burk plot:

- A) Change in y-intercept
- **B) Common V_{max} ****
- C) Change in slope only
- D) Parallel lines

20. Gibbs free energy helps determine if reaction is:

****A) Thermodynamically feasible****

B) Kinetically fast

C) Under kinetic control

D) Under product inhibition

21. Which statement about EPR spectroscopy is CORRECT?

****A) Based on absorption bands of metal ion transitions****

B) Used for characterizing free radical intermediates

C) Provides information on vibrational states

D) Gives fragmentation pattern in mass spectrum

22. Which photorespiratory pathway enzyme is different in C4 plants?

A) Glyoxylate aminotransferase

****B) Glycolate oxidase****

C) Alanine aminotransferase

D) Malate synthase

23. Identify role of sodium dodecyl sulphate in gel electrophoresis:

****A) Negatively charges and denatures proteins****

B) Antioxidant prevents diffusion

C) Binds to proteins reversibly

D) Deionizes water in apparatus

24. DNA replication in E.Coli is carried out by DNA polymerase _____:

A) I

****B) III****

C) II

D) IV

25. Which statement about convex lens is CORRECT?

A) Only one focal point

****B) Images can be real or virtual****

C) Suffers from chromatic aberration

D) Acts as an electromagnetic radiation source

Here are 25 fill in the blanks questions from the protein, carbohydrate and lipid tests sections of the biology lab manual:

1. The _____ test is used to detect the presence of proteins.

2. Formation of brick-red precipitate in _____ test indicates presence of reducing sugars.
3. Iodine solution results in _____ color with starch but not with monosaccharides.
4. Sulfuric acid is added in _____ test to catalyze dehydration of sugars.
5. A fat solvent mixture can be made using _____ and _____.
6. _____ is an enzyme found in saliva that catalyzes breakdown of starch.
7. _____ structure refers to folding of a polypeptide chain into units like alpha helices.
8. _____ is defined as number of milligrams of KOH required to neutralize fatty acids obtained from 1 gram fat.
9. In paper chromatography, the retardation factor R_f indicates relative _____ of solutes.
10. The distance moved by the biomolecule in chromatography is divided by the distance moved by the _____ to get R_f value.

11. Ninhydrin test detects _____ group present in amino acids which reacts to give purple color.

12. Formation of soap and _____ is the indication of a positive saponification test.

13. Xanthoproteic test gives yellow coloration due to reaction of proteins containing _____ amino acids with nitric acid.

14. Iodine forms a colored complex with _____ polysaccharides like starch but not others like cellulose.

15. Benedict's test detects reducing sugars having a free _____ group.

16. Biuret test is used to detect the presence of _____ containing two or more peptide bonds.

17. The separation of compounds by paper chromatography depends on the _____ properties of the amino acids.

18. _____ structure refers to the sequence in which amino acids are linked in a polypeptide chain.

19. In chromatography, the stationary phase refers to the _____ material.

20. The saponification value helps determine average fatty acid _____ length.
21. Enzyme activity is optimum at a particular pH called the _____ pH.
22. Molisch's test confirms presence of _____ in a sample by formation of a complex with furfurals.
23. Tyrosine and tryptophan contain _____ ring systems detected by xanthoproteic test.
24. In Biuret test, violet coloration is due to complex formation between _____ ions and peptides.
25. Solubility test can differentiate _____ which dissolve in organic solvents from carbohydrates.

Answers:

1. Biuret
2. Benedict's
3. Blue-black
4. Molisch's
5. Ethanol, ether
6. Amylase

7. Secondary
8. Saponification value
9. Mobility
10. Solvent
11. Amino
12. Glycerol
13. Aromatic
14. Branched
15. Aldehyde
16. Proteins
17. Side chain
18. Primary
19. Filter paper
20. Chain
21. Optimum
22. Carbohydrates
23. Benzene
24. Copper
25. Lipids